

University of Dhaka
Affiliated Engineering Colleges
B. Sc. in Computer Science & Engineering
2nd Year 1st Semester Final Examination-2021
MATH 2105 (Linear Algebra)

Time: 3.00 Hours

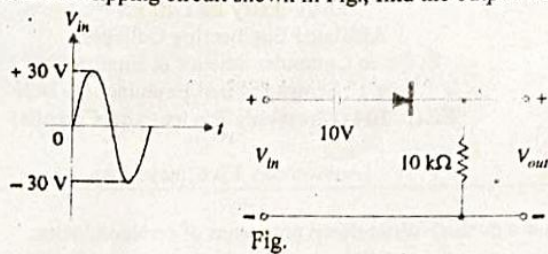
Marks: 70

[Answer any Five questions]

1.
 - a) What is periodic matrix? Find the periodic of the matrix $\begin{pmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{pmatrix}$ 4
 - b) Solve the system of equations by Cromer's rule: $2x - y + 3z = -2$ 4
 $x + 2y - z = 9$
 $3x + 2y + 3z = 9$
 - c) Define minors and cofactors. Prove that, $\begin{vmatrix} \lambda+a & b & c & d \\ a & \lambda+b & c & d \\ a & b & \lambda+c & d \\ a & b & c & \lambda+d \end{vmatrix} = \lambda^3(\lambda+a+b+c+d)$ 6
2.
 - a) Define inverse matrix. Prove that $(AB)^{-1} = B^{-1}A^{-1}$, where A & B are non-singular matrix. 5
 - b) Define rank of a matrix. Find the rank of the matrix $\begin{pmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & 2 & 5 \end{pmatrix}$ 5
 - c) Solve the system of linear equation: $2x - 3y + 5z = 1$, $3x + y - z = 2$, $x + 4y - 6z = 1$ 4
3.
 - a) State and prove Cayley-Hamilton theorem. 1+5
 - b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. Find A^{-1} using this theorem. 5+3
4.
 - a) Apply Gram-Schmidt Process to transform the basis $u_1 = (1,1,1)$, $u_2 = (1,1,0)$ and $u_3 = (1,0,0)$ into an orthogonal basis. 7
 - b) Show that every square matrix can be uniquely expressed as the sum of a symmetric and skew symmetric matrix. 5
 - c) Define kernel and range of a linear Transformation. 2
5.
 - a) What is linear transformation? Prove that $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T(x, y, z) = (2x + y + z, 3x + 2y + 4z)$ is linear. 1+3
 - b) Define linear dependence and independence of vectors. Show that the vectors $(1,0,1)$, $(-3,2,6)$ and $(4,5,-2)$ are linearly independence. 2+2
 - c) If $T: V(F) \rightarrow V(F)$ be a linear operator and $v_1, v_2, \dots, v_n$ are Eigen vectors belonging to distinct Eigen values $\lambda_1, \lambda_2, \dots, \lambda_n$ respectively then show that the vectors are linearly independent. 6
6.
 - a) Define direct sum of a vector space. Show that the vector space V is the direct sum of it's subspaces S and T iff (i). $V = S + T$ (ii). $S \cap T = \{0\}$ 8
 - b) Define diagonal matrix. Diagonalize the matrix $A = \begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix}$ and find an invertible 2×2 matrix such that $P^{-1}AP$ is a diagonal matrix with the eigenvalues of A down the diagonal. 6
7.
 - a) What is matrix Polynomial? Let $f(t) = t^2 - 4t + 7$ then prove that $f(A) = 0$ where $A = \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix}$. Also find the value of A^3 . 7
 - b) Determine the value of λ such that the following system of linear equation has (i) no solution (ii) more than one solution (iii) a unique solution $x + y + \lambda z = 1$, $x + \lambda y + z = 1$, $\lambda x + y + z = 1$ 7

7. a) For the input wave to the clipping circuit shown in Fig., find the output waveform.

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- b) For the input waveform to the clipping circuit in Fig., find the output voltage waveform.

7

